Government Engineering College, Bhavnagar
Computer Engineering Department
B.E. C.E. – V Division: E
Subject: Computer Networks (BE05000171)
Term: A.Y. 2026-27 (Odd) Duration: 17-Jun-2026 to 14-Oct-2026
(Societal Internship: 17-Jun-2026 to 30-Jun-2026)

PBL ACTIVITIES 2 AND 3 • TECHNICAL WRITING AND CASE STUDY

Research Article and Case Study — Combined Instruction Sheet

Activity 2: Research article on 5G, SDN or cybersecurity | Activity 3: Case study on a cyberattack or a network outage

PBL CATEGORIES	ALLOTTED HOURS	MODE OF WORK	DELIVERABLES	LAST DATE OF SUBMISSION
Technical Writing; Case Study	**10 + 10 Hours**	**Individual**	**Report; Report + PPT Handout**	**25-09-2026**

0 How Your Topics Are Decided

Topics are assigned from your enrolment number so that no selection, negotiation or first-come basis is involved. Take the **last two digits** of your enrolment number and call them **X** and **Y**:

X = the second-last digit **Y** = the last digit

Activity	First selector	Second selector
Activity 2 — Research Article	Topic number = **Y** (Table 2.1)	Focus dimension = **X** (Table 2.2)
Activity 3 — Case Study	Incident number = **X** (Table 3.1)	Analysis dimension = **Y** (Table 3.2)

Worked example. For enrolment number 23 06 91 07 **4 2**, X = 4 and Y = 2. That student writes Activity 2 on *Topic 2* viewed through *Focus dimension 4*, and Activity 3 on *Incident 4* analysed through *Analysis dimension 2*.

The two selectors combine to give one hundred distinct assignments per activity. Two students may share a broad topic, but they will have been given different dimensions to write it through, and the submissions must therefore differ substantially. Both the topic number and the dimension number must be printed on the title page of your submission — a submission that does not match the writer's enrolment number will not be evaluated.

1 Activity 2 — Research Article on 5G, SDN or Cybersecurity

Objective. To locate, read and synthesise current technical literature on an emerging networking topic and to communicate it in the form of a short, well-referenced technical article written in the student's own words. The activity addresses **CO-4** and **CO-5** at the *Analyse* and *Evaluate* levels.

Table 2.1 Broad Topic — selected by the last digit (Y)

Y	Topic	Indicative coverage
0	**5G network architecture**	The 5G radio access network, the 5G core (5GC), the service-based architecture, and network slicing.
1	**Edge computing and URLLC in 5G**	Multi-access edge computing, ultra-reliable low-latency communication, and the classes of application that depend on them.
2	**Software Defined Networking**	Separation of the control and data planes, the OpenFlow protocol, flow tables and the SDN controller.
3	**Network Function Virtualization**	Virtualised network functions, the NFV MANO framework, and service function chaining.
4	**SDN controllers and programmability**	Comparison of controller platforms, northbound and southbound interfaces, and intent-based networking.
5	**Zero-trust network architecture**	The "never trust, always verify" model, micro-segmentation, identity-based access and continuous verification.
6	**Machine learning for intrusion detection**	Signature-based versus anomaly-based detection, feature extraction from network traffic, and the false-positive problem.
7	**Post-quantum cryptography in networks**	The threat to current public-key cryptography, the standardised post-quantum algorithms, and their effect on TLS and VPN.
8	**Ransomware and network-level defence**	The ransomware attack lifecycle, lateral movement inside a network, and the controls that contain it.
9	**AI-driven network monitoring**	Telemetry collection, anomaly detection on network data, automated root-cause analysis and self-healing networks.

Table 2.2 Focus Dimension — selected by the second-last digit (X)

Your article must be written **through this lens**. The lens determines what the main body of the article argues; it is not an additional section bolted on at the end.

X	Focus dimension	What the main body must therefore address
0	**Architecture and protocol operation**	How the technology actually works: its components, the protocol exchanges between them, and the sequence of operations.
1	**Performance and scalability**	Reported throughput, latency, capacity and scaling limits, with quantitative figures from published sources.
2	**Security vulnerabilities and countermeasures**	The attack surface the technology introduces, documented weaknesses, and the mitigations proposed for them.
3	**Standardisation status**	Which body governs it (3GPP, IETF, ONF, NIST, IEEE), which releases or RFCs define it, and what remains unstandardised.
4	**Deployment in India**	Where and by whom it has actually been deployed in India, at what scale, and what the reported outcomes were.

X	Focus dimension	What the main body must therefore address
5	**Cost, energy and sustainability**	Capital and operating cost implications, energy consumption, and the sustainability argument for or against it.
6	**Comparison with what it replaces**	A structured comparison against the preceding technology, stating clearly what improved and what became harder.
7	**Open-source tools and testbeds**	The tools, simulators and testbeds available to experiment with the technology, and what each of them can and cannot demonstrate.
8	**Interoperability and migration**	How the technology coexists with legacy infrastructure, and the practical difficulties of migrating to it.
9	**Open research problems**	What remains unsolved, which problems the literature identifies as hard, and the likely direction over the next five years.

1.1 Required Structure of the Article

#	Section	Requirement
1	Title page	Article title framed by the student, student name, enrolment number, **topic number Y and focus dimension X**, division/batch, semester, faculty name.
2	Abstract	100 to 150 words summarising what the article covers and what it concludes.
3	Introduction	Context, why the topic matters, and a statement of what the article will examine through the assigned focus dimension.
4	Technical background	The concepts and terminology a reader needs before the main body; approximately one fifth of the article.
5	Main body	The substance of the article, written through the assigned focus dimension; approximately half of the article. This is the section that distinguishes your submission from another student's on the same topic.
6	Discussion	The student's own assessment: what the evidence supports, where sources disagree, and what the practical implication is.
7	Conclusion	Three to five sentences. No new material.
8	References	Minimum **five** sources in IEEE style, numbered and cited in the text as [1], [2] and so on.

1.2 Format and Length

- **1200 to 1500 words** excluding the title page, references and figure captions. State the word count on the title page.
- A4, 11 pt body text, 1.15 line spacing, justified, 1-inch margins, numbered headings, page numbers.
- At least **one original figure** drawn by the student — an architecture diagram, a protocol exchange, a comparison chart or a taxonomy. A figure copied from a source is permitted *in addition*, with the source credited in the caption, but does not satisfy this requirement.
- At least **one table** that organises information rather than merely listing it.
- Submitted as a single PDF named `<EnrolmentNo>_Activity2.pdf`.

1.3 Assessment Criteria

In line with the syllabus criteria for this PBL category — *originality and technical content*:

- **Fidelity to the assignment** — whether the article genuinely addresses the assigned focus dimension rather than a general survey of the topic.
- **Technical accuracy and depth** — correctness of the concepts, and whether the treatment goes beyond definitions.

- **Originality of expression** — whether the article is written in the student's own words and reasoning.
- **Quality of sources** — whether standards documents, papers and vendor technical material were used rather than only blog posts.
- **Figure and table** — whether the original figure adds understanding.
- **Presentation of the document** — structure, referencing discipline, language and formatting.
- **Oral discussion during term work submission** — the student will be questioned on the article at the time of term work submission: the concepts used, the figure drawn, the sources cited, and why the assigned focus dimension leads to the conclusion reached.

2 Activity 3 — Case Study on a Cyberattack or a Network Outage

Objective. To reconstruct and analyse a documented real-world incident, identify its technical root cause, trace how the failure or compromise propagated through the network, and derive defensible recommendations. The activity addresses **CO-4** and **CO-5** at the *Analyse* and *Evaluate* levels.

Table 3.1 Incident — selected by the second-last digit (X)

X	Incident	Nature of the incident
0	**Colonial Pipeline ransomware attack** United States, May 2021	Ransomware intrusion through a legacy VPN account, leading to the shutdown of a major fuel pipeline.
1	**Facebook / Meta global outage** 4 October 2021	A configuration change withdrew the organisation's BGP routes, making its entire address space unreachable for hours.
2	**WannaCry ransomware worm** May 2017	Self-propagating ransomware exploiting an SMBv1 vulnerability, spreading across networks worldwide.
3	**Mirai botnet and the Dyn DNS attack** October 2016	Compromised IoT devices used to launch a distributed denial-of-service attack against a major DNS provider.
4	**SolarWinds Orion supply-chain compromise** Disclosed December 2020	Malicious code inserted into a trusted network-management product's update mechanism.
5	**AIIMS New Delhi ransomware attack** India, November 2022	Ransomware disabling hospital information systems at a major public hospital.
6	**Rogers Communications nationwide outage** Canada, 8 July 2022	A routing configuration change that removed a national carrier's network from service for most of a day.
7	**Log4Shell mass exploitation** CVE-2021-44228, December 2021	A remote code execution flaw in a widely embedded logging library, exploited at Internet scale within days.
8	**Cloudflare global outage** 2 July 2019	A deployed rule caused CPU exhaustion across the provider's edge, taking a large fraction of its customers offline.
9	**Change Healthcare ransomware attack** United States, February 2024	Ransomware against a healthcare payments processor, disrupting claims and prescription services nationally.

Table 3.2 Analysis Dimension — selected by the last digit (Y)

Your case study must be organised so that this dimension forms the **core analytical section** of the report.

Y	Analysis dimension	What the core section must therefore establish
0	**Reconstructed timeline**	An hour-by-hour or day-by-day timeline from initial access or first fault to full restoration, every entry sourced.
1	**Protocols and layers involved**	Which OSI or TCP/IP layers and which specific protocols were implicated at each stage, and why those protocols behaved as they did.
2	**Technical root cause**	The precise technical failure — the misconfiguration, the vulnerable component, the design assumption that did not hold.
3	**Blast radius**	Who and what was affected downstream, how the dependency chain carried the failure outward, and the measurable scale of impact.
4	**Detection**	How the incident was first noticed, how long it went unnoticed, and what monitoring would have surfaced it earlier.

Y	Analysis dimension	What the core section must therefore establish
5	**Containment and recovery**	The actions taken to stop the spread and restore service, in order, and why recovery took as long as it did.
6	**Preventive controls**	Which specific technical controls, had they been present, would have prevented or limited the incident — argued against the actual root cause, not listed generically.
7	**Regulatory and disclosure aftermath**	Notification obligations, regulatory response, penalties, and how disclosure was handled.
8	**Comparative analysis**	A structured comparison with any one other incident from Table 3.1, identifying what the two share and where they diverge.
9	**Applicability to an Indian campus or enterprise network**	Whether an equivalent failure could occur in a network of the scale of your own institute or a mid-sized Indian enterprise, and what would have to be true for it to happen.

2.1 Required Structure of the Report

#	Section	Requirement
1	Title page	Report title, student name, enrolment number, **incident number X and analysis dimension Y**, division/batch, semester, faculty name.
2	Incident at a glance	A fact box: organisation, sector, date, duration, incident type, and a one-line statement of impact.
3	Background	What the affected organisation does, and what its network or service was expected to deliver.
4	What happened	A factual narrative of the incident, with sources cited for each claim.
5	Core analysis	The assigned analysis dimension, treated in depth; approximately **40 percent** of the report. This is the section that distinguishes your submission from another student's on the same incident.
6	Impact	Technical, operational, financial and human consequences, quantified wherever a figure is available.
7	Lessons and recommendations	At least **four** recommendations, each tied to a specific finding in your own analysis and each naming the mechanism — not "improve security", but which control, applied where.
8	Your own assessment	One paragraph stating which single decision or omission the student judges most responsible, and why.
9	References	Minimum **five** sources in IEEE style. At least two must be primary — an official post-mortem, an advisory from a national CERT, a regulatory filing or a vendor technical bulletin.

2.2 Format and Length

- **1200 to 1500 words** excluding the title page, references and figure captions. State the word count on the title page.
- A4, 11 pt body text, 1.15 line spacing, justified, 1-inch margins, numbered headings, page numbers.
- At least **one original diagram** drawn by the student showing either the attack path or the propagation of the failure through the affected infrastructure.
- At least **one table** — a timeline, a comparison, or an impact summary.
- Submitted as a single PDF named `<EnrolmentNo>_Activity3.pdf`.

2.3 PPT Handout

- A slide handout of **8 to 10 slides**, submitted as a PDF exported in **handout format** (2 or 3 slides per page, with the header showing the student's name and enrolment number).
- Suggested sequence: title slide → incident at a glance → background → what happened → two to three slides on the assigned analysis dimension → impact → recommendations → the student's own assessment.
- The original diagram from the report must appear on one of the slides.
- Slides must use figures, timelines and short points, not paragraphs lifted from the report.
- The handout is merged into the same PDF as the report, as described in Section 3.4 of this sheet; a submission with only one of the two is incomplete.

2.4 Assessment Criteria

In line with the syllabus criteria for this PBL category — *analysis and reporting*:

- **Depth of the core analysis** — whether the assigned dimension is genuinely investigated rather than summarised.
- **Technical correctness** — whether the protocols, mechanisms and root cause are described accurately.
- **Evidence discipline** — whether claims are sourced, and whether primary sources were used.
- **Quality of recommendations** — whether each recommendation follows from a finding in the student's own analysis.
- **Independent judgement** — the reasoning in the student's own assessment.
- **Report and handout quality** — structure, the original diagram, language and formatting.
- **Oral discussion during term work submission** — the student will be questioned on the case study at the time of term work submission: the sequence of events, the technical root cause, the diagram drawn, and the reasoning behind each recommendation made.

3 Rules Applying to Both Activities

3.1 Originality

- Both submissions must be written by the student. Text reproduced from any source without quotation and citation is plagiarism and the submission will be rejected.
- Text generated by an AI writing tool and submitted as the student's own work is treated in the same way as plagiarism.
- Every submission must end with the following signed declaration:

  "I declare that this submission is my own work, that all material drawn from other sources is cited, and that I am able to explain every statement it contains."
- The student must retain the sources consulted and be able to produce them on request.
- The student may be asked, at the time of term work submission, to explain any part of the submission — a definition used, a figure drawn, or a recommendation made. Inability to do so will be treated as evidence that the work is not the student's own.

3.2 Reference Table

The reference list of each submission must be presented in the following form, so that sources can be verified:

No.	Author / Organisation	Title	Type	Year	Accessed on
1	e.g. 3GPP	TS 23.501 System architecture for the 5G System	Standard	2023	DD-MM-YYYY
2					
3					

Type: Standard / RFC / Journal paper / Conference paper / Advisory / Post-mortem / Vendor documentation / News report. The full URL must accompany each entry in the IEEE-style reference list itself.

3.3 General Instructions

- Print the topic/incident number and the dimension number on the title page of every submission, along with the last two digits of the enrolment number used to derive them.
- An assignment cannot be exchanged with another student or changed on request.
- Word limits are enforced in both directions. A submission substantially below or above the range will be returned.
- Screenshots of web pages are not a substitute for the student's own figures.
- Submit the PDF files with the exact file names specified. Files submitted in any other format will not be evaluated.
- Two submissions found to be substantially similar will both be rejected, regardless of who wrote first.

3.4 Form of Submission — applies to both activities

Each activity must be submitted as one single PDF file. Where an activity has more than one component — the Activity 3 report and its PPT handout — both components must be merged into that same PDF, the report first and the handout after it. Separate files, ZIP archives, Word documents, images or printouts of individual sections will not be accepted.

Name the files exactly as follows, using your own enrolment number:
`<EnrolmentNo>_Activity2.pdf` and `<EnrolmentNo>_Activity3.pdf`
A file submitted under any other name, or in any other format, will not be evaluated.

Faculty in charge: _____ Date of issue: _____ Last date of submission (both activities): **25-09-2026**